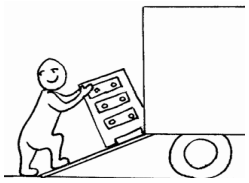


Inclined Planes Mini-Assessment



Use Pearson Science p.301 - 305 TOTAL: _____ 39

1. **List** three places you have used a ramp.

   a.

   _____
   _____

   b.

   _____
   _____

   c.

   _____
   _____

   (3 marks)

2. **List** three common examples of a wedge.

   a. _____
   _____

   b. _____
   _____

   c. _____
   _____

   (3 marks)

3. **State** whether the following are true or false.

   a. The longer the slope of a ramp for a certain height, the greater its mechanical advantage.

   _____

b. A screw with a small pitch corresponds to a steep inclined plane.

_____

c. A screw with a small pitch requires fewer turns to be inserted into a piece of wood than a screw with a large pitch.

_____

d. A wedge changes the direction of a force by 90° and increases its size.

_____ (4 marks)

4. **State** an example of a type of screw that cuts through a:

a) solid

_____

b) liquid

_____

c) gas

_____

(3 marks)

5. **Recall** five devices that can be produced by flaking.

_____

_____

_____

_____

(5 marks)

6. **Recall** one name for a spear thrower

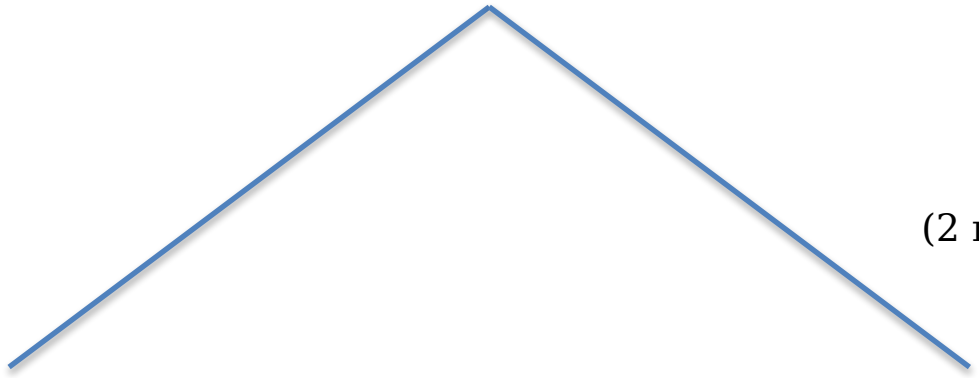
_____

(1 mark)

7. You have been given the job of building a road up the following hill.

a) **Draw** in the road that would be shortest but hardest to walk up.

b) **Draw** in a road that would be long but easy to walk up.



(2 marks)

8. **Describe** how a stone is flaked

_____

_____

_____

_____

(2 marks)

9. **Explain** why stone rich in silica is suitable to use in creating stone tools

_____

_____

_____

_____

_____
_____

(1 mark)

10. **Identify** which type of simple machine (inclined plane, wedge or screw) is described by these devices:

    a) corkscrew _____

    b) axe _____

    c) electric fan _____

    d) car park ramp _____

    e) chisel _____

    f) escalator _____

(6 marks)

11. Mechanical advantage is equal to $\frac{\text{length of slope}}{\text{height of ramp}}$

    Calculate the mechanical advantage of the following ramps:

Length of Slope (m)	**Height of Ramp (m)**	**Mechanical Advantage**
15	3	
50	25	
12	2	
20	5	
240	30	

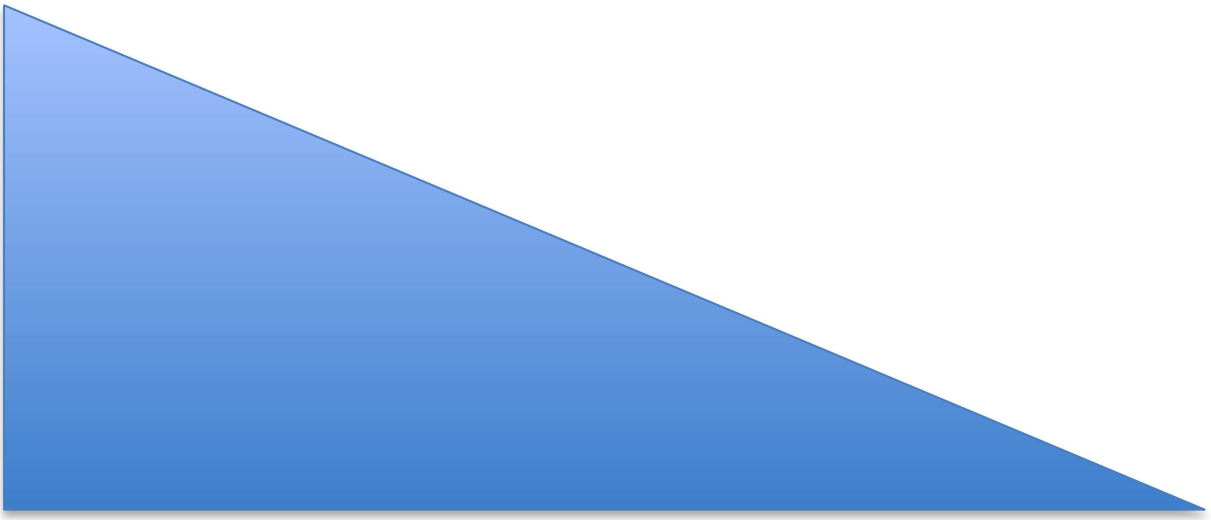
(5 marks)

12. Use a piece of scrap paper to cut out a gentle slope and a steep slope (see p.304 SB) and wrap them around a pencil.

Gentle Slope



Steep Slope



When you wrapped them around a pencil and produced a "screw", was the thread spread out or tight together?

a) Gentle slope

_____
_____

b) Steep slope

_____
_____

(2 marks)

13. **Draw** an axe head and show how it is two inclined planes.

(2 marks)